

Cauchy Distribution:

A random variable x is said to be followed a cauchy distribution with parameter p and n if its probability density function (pdf) is given by;

$$f(x) = \frac{1}{\pi} \frac{1}{[r^2 + (x-\lambda)^2]}, \quad -\infty < x < \infty$$

where, $\lambda > 0, r > 0$

$x \sim C(\lambda, r)$

scale parameters

$\rightarrow$ location parameter

Proof:

In ΔABC ,

$$\tan \theta = \frac{x-\lambda}{r}$$

Differentiating,

$$\sec^2 \theta d\theta = \frac{dx}{r}$$

$$\Rightarrow d\theta = \frac{dx}{r \sec^2 \theta}$$

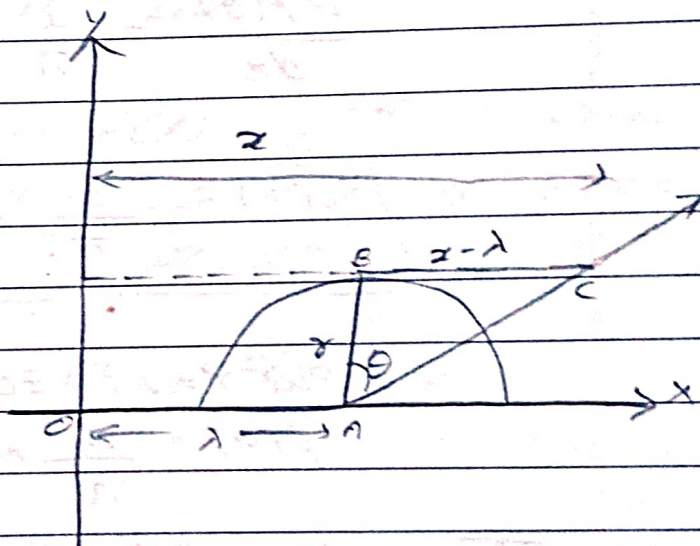
$$\Rightarrow d\theta = \frac{dx}{r(1 + \tan^2 \theta)}$$

$$\Rightarrow d\theta = \frac{dx}{r \left[1 + \frac{(x-\lambda)^2}{r^2} \right]}$$

$$\Rightarrow d\theta = \frac{r dx}{r^2 + (x-\lambda)^2}$$

Again, for probability density function,

$$P(\theta) = \frac{\text{favourable no. of cases}}{\text{total no. of cases}}$$



$$p(\theta) = \frac{\theta}{\pi}$$

$$\boxed{p(\theta) = \frac{\theta}{\pi}}$$

Again, differentiating,

$$dp(\theta) = \frac{d\theta}{\pi}$$

$$dp(\theta) = \frac{1}{\pi} \times \frac{r dx}{r^2 + (x-\lambda)^2}$$

$$\frac{dp(\theta)}{dx} = \frac{1}{\pi} \times \frac{r}{[r^2 + (x-\lambda)^2]}$$

$$\boxed{f(x) = \frac{1}{\pi} \times \frac{r}{[r^2 + (x-\lambda)^2]}}$$

Put $r=1$, $\lambda=0$,

$$f(x) = \frac{1}{\pi} \times \frac{1}{(1+x)^2}$$

This is the pdf of standard Cauchy distribution
That $x \in C(0,1)$.

λ, γ — scale
— location

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✓ Characteristics function of Cauchy distribution

The characteristics function of a r.v. X is denoted by $\phi_X(t)$ and defined

$$\phi_X(t) = E(e^{itx})$$

Here $X \sim C(0, 1)$ then its pdf is

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}, \quad -\infty < x < \infty$$

Now,

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{\pi} \cdot \frac{1}{1+x^2} dx$$

Consider a function $f(t) = e^{-a|t|}$ then Fourier transformation of $f(t)$ is given by

$$F(x) = \int_{-\infty}^{\infty} e^{-itx} f(t) dt$$

$$= \int_{-\infty}^{\infty} e^{-itx} e^{-a|t|} dt$$

$$= \int_{-\infty}^0 e^{-itx} e^{at} dt + \int_0^{\infty} e^{-itx} e^{-at} dt$$

$$= \int_{-\infty}^0 e^{(a-ix)t} dt + \int_0^{\infty} e^{-(a+ix)t} dt$$

$$= \left[\frac{e^{(a-ix)t}}{(a-ix)} \right]_{-\infty}^0 - \left[\frac{e^{-(a+ix)t}}{(a+ix)} \right]_0^{\infty}$$

$$= \frac{1}{a-ix} + \frac{1}{a+ix}$$

$$F(x) = \frac{2a}{a^2 - i^2 x^2} = \frac{2a}{a^2 + x^2}$$

$$\frac{df(x)}{dx} = f(x)$$

Now, the Inverse transformation of $F(x)$ is

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(x) e^{itx} dx$$

$$\text{or, } e^{-a|t|} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2a}{x^2 + a^2} e^{itx} dx$$

$$\text{or, } \frac{e^{-a|t|}}{a} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itx}}{x^2 + a^2} dx$$

Take $a=1$,

$$e^{-|t|} = \left[\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2} dx \right]$$

$$e^{-|t|} = \phi_x(t)$$

$$\therefore \phi_x(t) = e^{-|t|}$$

This is the required characteristic function of standard Cauchy distribution.

Theorem:

The arithmetic mean of a sample of n independent observations drawn from a standard Cauchy distribution is also a standard Cauchy distribution.

Proof:

Let $x_1, x_2, \dots, x_n$ be a random sample of size n drawn from a standard Cauchy distribution. Then,

$$x_i \sim C(0,1); i=1,2,\dots,n$$

Then, the sample mean is given by

$$\bar{x} = \frac{\sum x_i}{n}$$

We have,

characteristic function of X_i is;

$$\phi_{X_i}(t) = e^{-|t|}$$

Then, the characteristic function of $\bar{X}$ is given by;

$$\phi_{\bar{X}}(t) = \phi_{\sum X_i}(t/n)$$

$$= \prod_{i=1}^n e^{-|t/n|}$$

$$\phi_{\bar{X}}(t) = \left[e^{-|t/n|} \right]^n$$

$$\phi_{\bar{X}}(t) = e^{-|t|}$$

Thus, the result followed by uniqueness theorem of characteristic function.

Hence, if $X_1, X_2, \dots, X_n$ is a random sample of size n drawn from a standard Cauchy distribution $C(0,1)$ then the sample mean ($\bar{X}$) is also standard Cauchy variate distributed as $C(0,1)$.

Properties of Cauchy distribution.

1. In Cauchy distribution, moment of any order do not exist. Thus mean and variance do not exist.
2. MGF of Cauchy distribution does not exist.
3. The characteristic function of Cauchy

distribution $C(\lambda, \gamma)$ is;

$$\phi_X(t) = e^{it\lambda - \gamma|t|}, \gamma > 0$$

4. The standard Cauchy distribution $C(0, 1)$ has lower quartile $Q_1 = -1$ and upper quartile $Q_3 = 1$.
5. Cauchy distribution follows additive property.
6. The arithmetic mean of a sample of n independent observation drawn from a standard Cauchy distribution is also a standard Cauchy distribution.

Example:

Let X be a r.v. with pdf

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{(1+x^2)}, -\infty < x < \infty$$

- (i) Determine $P(-1 < X < \sqrt{3})$
- (ii) Find the upper quartile for X .

Soln:

Given that $X \sim C(0, 1)$

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{(1+x^2)}, -\infty < x < \infty$$

- (i) The prob. that X lies between -1 and $\sqrt{3}$ is given by

$$P(-1 < X < \sqrt{3}) = \int_{-1}^{\sqrt{3}} f(x) dx$$

$$= \frac{1}{\pi} \int_{-1}^{\sqrt{3}} \frac{1}{(1+x^2)} dx$$

Upper Quartile = $3/4$

Lower Quartile = $1/4$

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$$= \frac{1}{\pi} \left[\tan^{-1} x \right]_{-1}^{\sqrt{3}}$$

$$= \frac{1}{\pi} \left[\tan^{-1} \sqrt{3} - \tan^{-1} (-1) \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi}{3} - \left(-\frac{\pi}{4} \right) \right]$$

$$= \frac{1}{3} + \frac{1}{4}$$

$$= \frac{7}{12}$$

(ii) The upper quartile Q_3 is given by;

$$\int_{-\infty}^{Q_3} f(x) dx = \frac{3}{4}$$

$$\text{or, } \frac{1}{\pi} \int_{-\infty}^{Q_3} \frac{1}{1+x^2} dx = \frac{3}{4}$$

$$\text{or, } \frac{1}{\pi} \left[\tan^{-1} x \right]_{-\infty}^{Q_3} = \frac{3}{4}$$

$$\text{or, } \frac{1}{\pi} \left[\tan^{-1} Q_3 + \frac{\pi}{2} \right] = \frac{3}{4}$$

$$\text{or, } \frac{1}{\pi} \tan^{-1} Q_3 + \frac{1}{2} = \frac{3}{4}$$

$$\tan^{-1} Q_3 = \frac{\pi}{4}$$

$$Q_3 = \tan^{-1} \pi/4$$

$$Q_3 = 1$$

Hence, for standard Cauchy distribution, upper quartile is equal to 1.

Example:

If x has a standard Cauchy distribution, find the pdf of x^2 and identify its distribution.

Soln:

Since x has a standard Cauchy distribution, the pdf of x is

$$f(x) = \frac{1}{\pi} \times \frac{1}{(1+x^2)}, \quad -\infty < x < \infty$$

The distribution function of $y = x^2$ is given by;

$$G(y) = P(Y \leq y) = P(x^2 \leq y)$$

$$= P(-\sqrt{y} < x < \sqrt{y})$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{\pi(1+x^2)} dx$$

$$= \frac{2}{\pi} \int_0^{\sqrt{y}} \frac{1}{1+x^2} dx$$

$$G(y) = \frac{2}{\pi} [\tan^{-1} x]_0^{\sqrt{y}}$$

$$G(y) = \frac{2}{\pi} \tan^{-1} \sqrt{y}$$

The pdf of y is given by;

$$g(y) = \frac{dG(y)}{dy} = \frac{2}{\pi} \frac{d}{dy} \tan^{-1} \sqrt{y}$$

$$= \frac{2}{\pi} \times \frac{1}{1+y} \times \frac{1}{2\sqrt{y}}$$

$$= \frac{1}{\pi} \frac{y^{-1/2}}{1+y}$$

$$= y^{1/2-1}$$

$$= \frac{\Gamma(1/2) \Gamma(1/2)}{\Gamma(1/2+1/2)} (1+y)^{-1/2-1/2}, \quad y > 0$$

which is the pdf of Beta distribution of second kind with parameter $(\frac{1}{2}, \frac{1}{2})$.

Thus $x^2 \sim B_2(\frac{1}{2}, \frac{1}{2})$.